Data Science

Lecture 04: Machine Learning

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Where We Left Off

In Lectures 02–03 *we* wrote the model down: the probabilities, the graph. From today the machine chooses the model from data — and it can only do that because we restrict what it may choose.

Write the model
Lectures 02–03

→

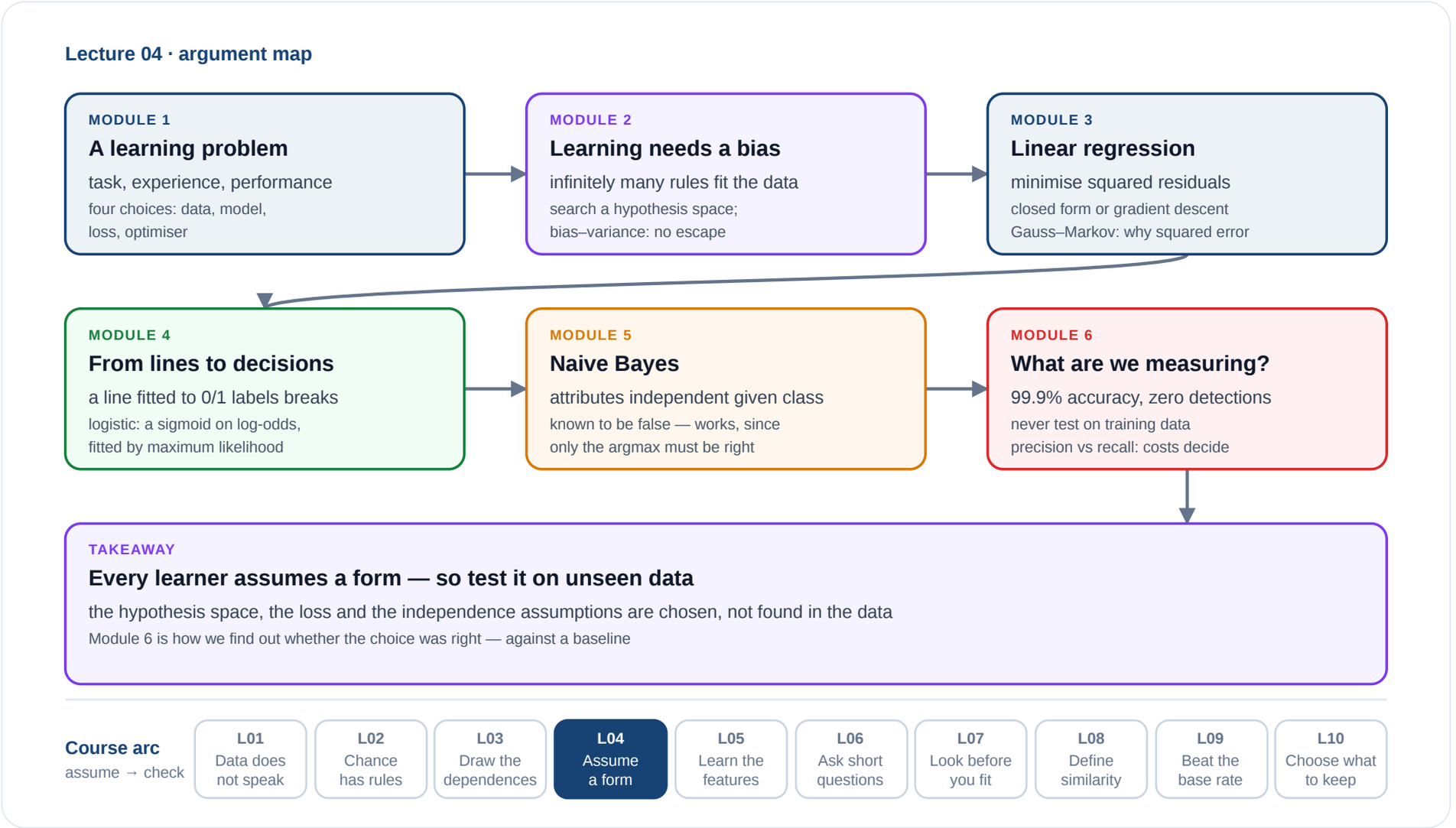
Learn it from data
Modules 1–5 — a form, a loss, an optimiser

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Check it on unseen data
Module 6

The base-rate trap of Lecture 02 returns in Module 6: a model with 99.9% accuracy that never detects a single case.

Argument Map



Module 1

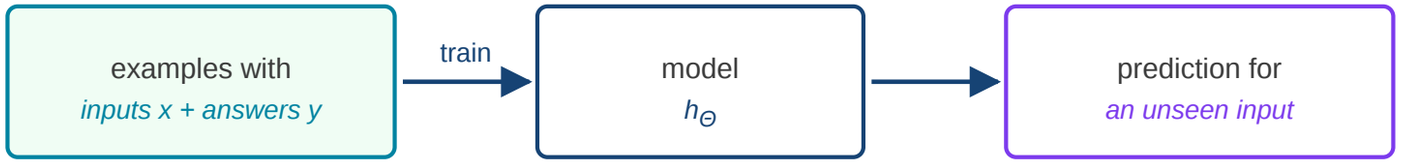
What Is a Learning Problem?

> ***Guiding Question:*** *What has to be true of a task — and of the data — before "learn it from examples" is a sensible answer?*

Learning From Answers

A radiologist has labelled ten thousand scans. A bank has ten years of loans and knows which ones defaulted. In both cases **every example carries its answer**.

Supervised: every training example carries the answer



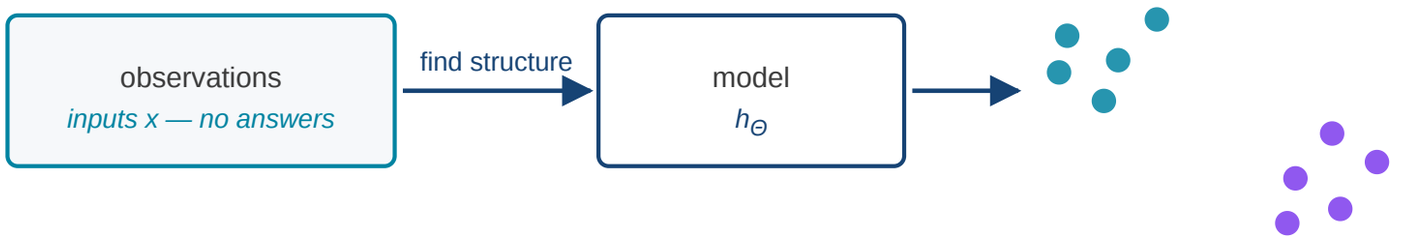
The learner is told the correct output for each case it sees.

What information does the learner receive during training?

Learning Without Answers

Now the same bank wants to know **what kinds of customer it has**. Nobody has ever written down the list — the categories are what we are looking for.

Unsupervised: no answers — only the observations themselves



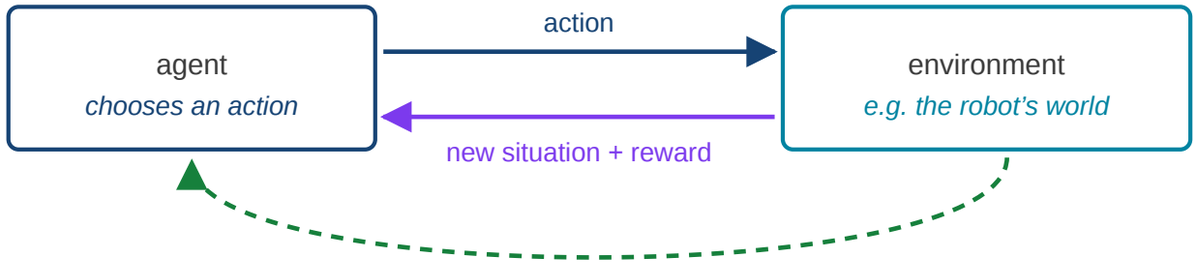
Structure has to be found, not reproduced.

What can we learn when nobody tells us the correct answer?

Learning by Acting

A robot has to cross a room. Nobody can label the correct motor command for every situation it might find itself in — and it only discovers whether a move was good **after making it**.

Reinforcement: no answer is given — only the consequence of acting



the consequence changes what the agent does next

The learner must act to find out whether the action was good.

How can a learner improve when it must act in order to find out whether the action was any good?

Same Goal, Different Information

Same goal, different information

	WHAT IT IS GIVEN	WHAT FEEDBACK IT GETS
Supervised	inputs + answers	the answer, every time
Unsupervised	inputs only	nothing
Reinforcement	the situation it acts in	a reward, after acting

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> The paradigms differ in **what the learner is given** and **what feedback it gets** — not in what we ultimately want from it.

What Is Common to All Three?

Nobody wrote the rule

The mapping from situation to answer is **too complex or too implicit** to specify by hand.

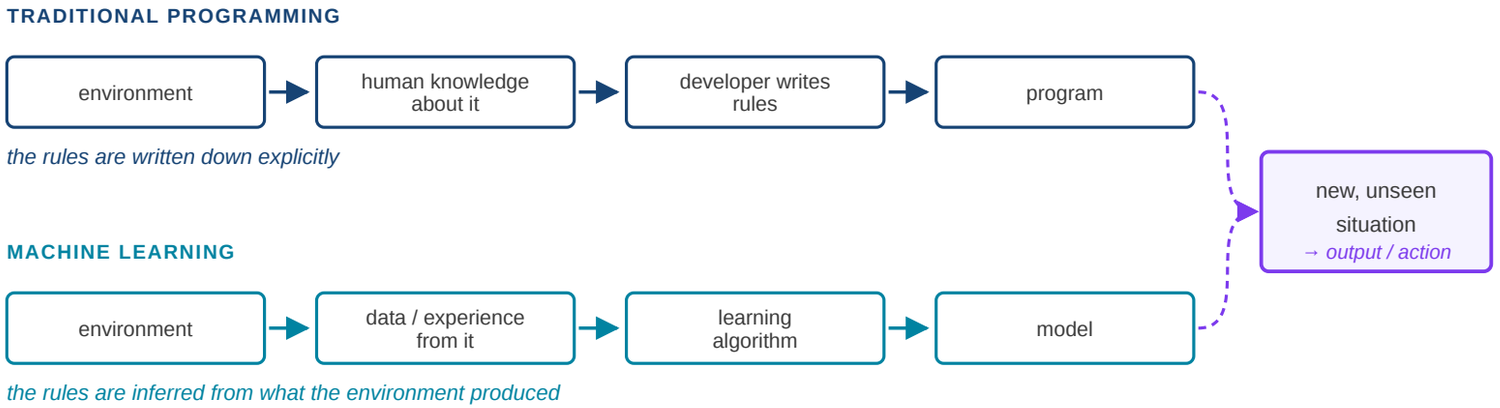
Examples exist

There is data — or experience — in which the desired behaviour is demonstrated, even if it was never described.

So what replaces the rule that nobody wrote?

Two Ways to Get Behaviour

Both are judged on situations neither of them has seen



What Actually Changed

TRADITIONAL PROGRAMMING

The developer's **knowledge becomes rules**. Deduction: rules plus input give the output.

The assumptions are visible — they are in the code.

MACHINE LEARNING

The environment's **data becomes a model**. Induction: examples plus a learning procedure give a rule.

The assumptions are still there — but now they are in the data, the model family and the objective.

The rules do not disappear when nobody writes them. They become **harder to inspect** — which is why the rest of this lecture is about making every part of that pipeline explicit.

What Would It Mean for a Machine to Learn?

Before any definition — three questions you would have to answer about *any* system before you could say it had learned anything:

- ? What is it supposed to get better at?
- ? What does it get to look at while getting better?
- ? How would we know whether it improved?

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Task T

what the system should learn to do

Experience E

what it learns from

Performance P

how we decide it improved

Chess

♟

A program that plays, loses, and plays again — millions of times against itself.

- → **Task** — play chess well
- → **Experience** — games played against itself, and positions from expert play
- → **Performance** — fraction of games won; rating against other engines

A Large Language Model

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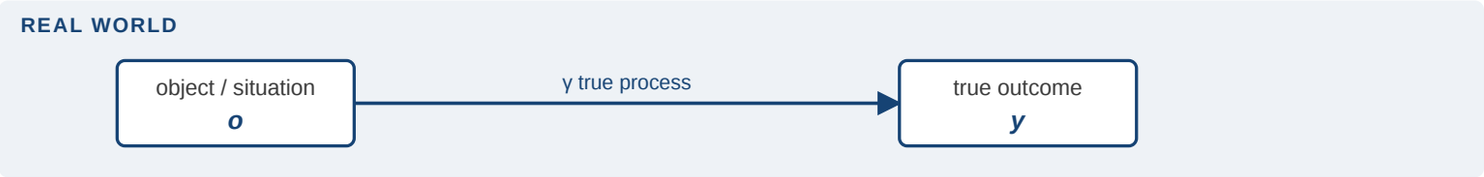
A model shown enormous quantities of text, asked only to continue it.

- → **Task** — predict the next word
- → **Experience** — a very large corpus of text, then human feedback
- → **Performance** — likelihood of held-out text; later, human judgement

Two systems with nothing in common — a board game and a text corpus — fit the **same three-part description**. That is what makes it a definition rather than a description of one program.

What Would We Ideally Like to Know?

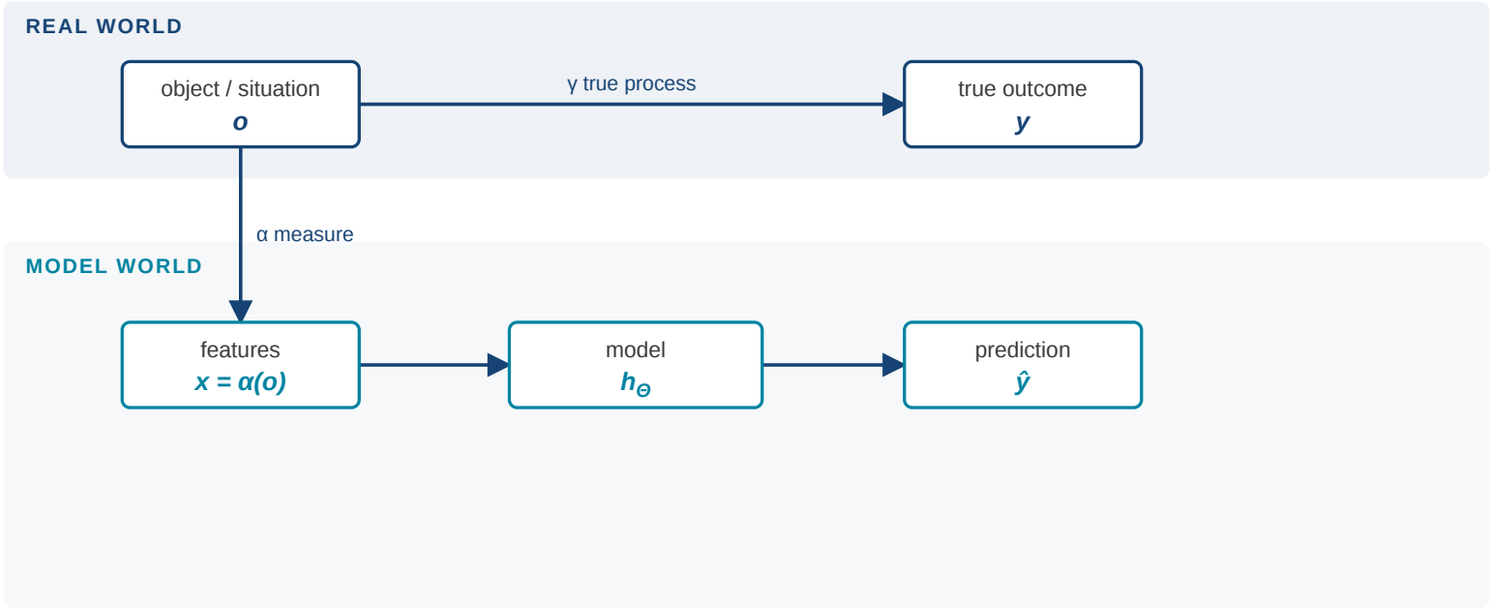
A process in the world turns situations into outcomes — we cannot see it directly



There is some real process that turns a situation into an outcome. Why can we not simply write it down?

From the Real World to a Model of It

The model maps features to a prediction



Two Worlds, One Correspondence

$$\gamma(o) \quad \rightsquigarrow \quad h_\Theta\big(\alpha(o)\big) \quad \equiv \quad h_\Theta(x)$$

γ

the **real process** — unknown, and not ours to write down

α

the **representation** — what we chose to measure

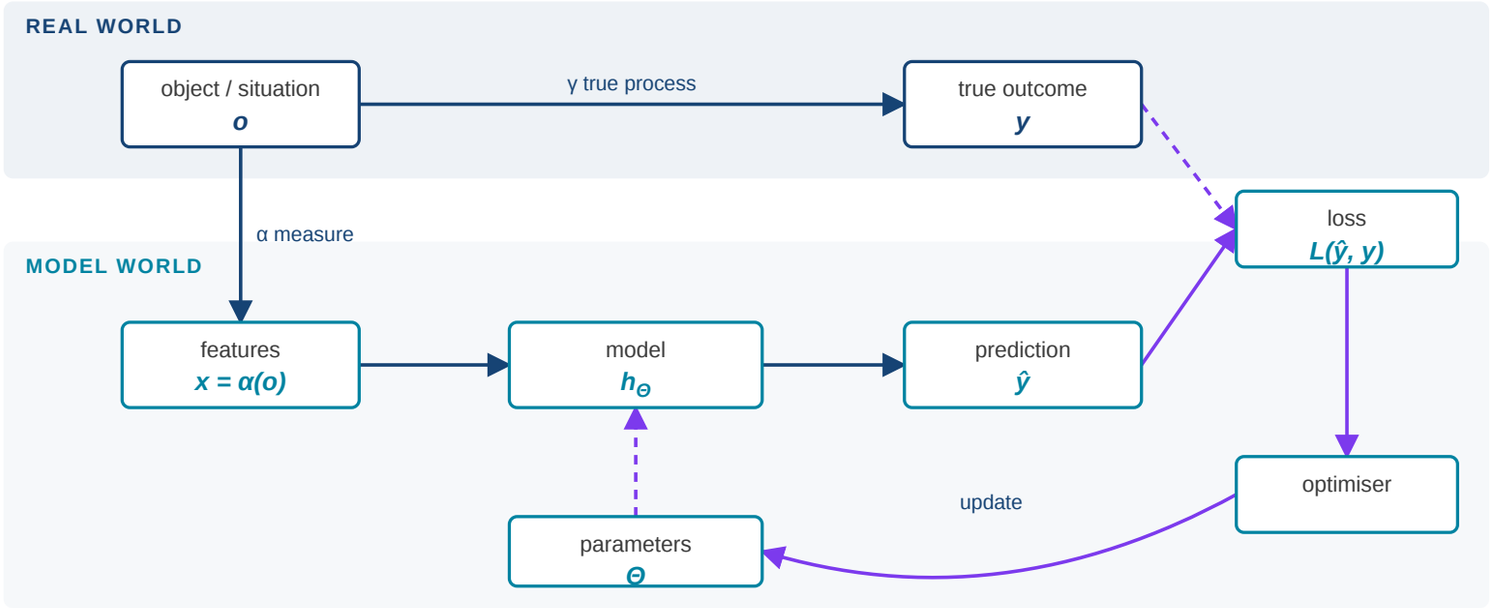
h_Θ

the **model** — our approximation, with parameters Θ

We never learn the world. We build a **model world** and approximate the process inside it.

Closing the Loop

The optimiser changes the parameters — and the training loop closes



The Four Choices

Data
what we measure

→

Model
h_Θ — what it may express

→

Loss
what counts as wrong

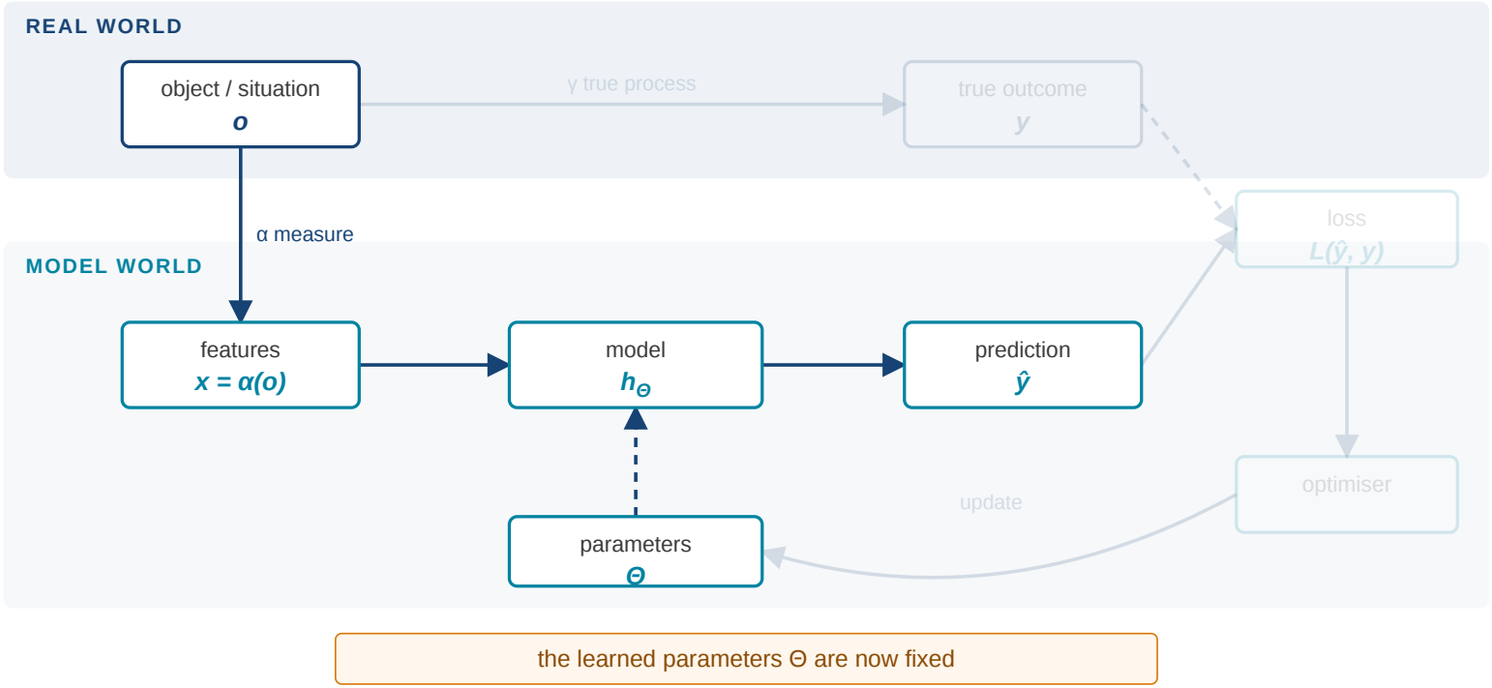
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Optimiser
how Θ is searched

The optimiser has settings of its own that are not learned from the data — **hyperparameters**. Choosing them is part of the outer process, not of the loop.

Training Is Not Using

Inference: a new, unseen situation follows one path — the training machinery is gone



the learned parameters Θ are now fixed

Which of these boxes still exist when the system is actually deployed?

The Task: Predicting Acceleration

The world

O — vehicles

$Y = \mathbb{R}$ — time to accelerate to 100 km/h

$\gamma(o)$ — the true acceleration time of vehicle o

The model world

α — measure wheels, horsepower, weight, torque

X — those four numbers, for every vehicle

h_Θ — a linear function of them

The Data

wheels	horsepower	weight	torque	acceleration time
4	150	1350	250	9.8
4	320	1600	430	4.9
6	95	2100	210	16.2
…	…	…	…	…

The last column is y; the rest is $\mathbf{x}$.



The Remaining Two Pieces

Loss function

Mean squared error — the average of the squared differences between predicted and observed acceleration times.

Learning algorithm

LMS — least mean squares, which adjusts the parameters to reduce that error.

A learning problem becomes concrete the moment you can name its **task, experience and performance measure**, and its system becomes buildable the moment you can name its four choices: **data, model, loss and optimiser**.

Designing the Learning Problem

Before any model is chosen, four things have to be decided — and each of them is a claim about the world. First, what the system is for:

Decide	Ask
The task	What should the system predict or decide? What is one unit of prediction? What counts as success?
The target	What is γ, in words a domain expert would accept? Is the label we have the thing we care about, or a proxy for it?

> A proxy label is the most consequential choice here: "was diagnosed" is not "had the disease", "was clicked" is not "was good".

Designing the Learning Problem — Data

THE DATA

- ? Which observations represent the problem?
- ? How were the examples sampled?
- ? Will future cases look like these?

THE REPRESENTATION A

- ? What are the inputs, concretely?
- ? Which features are available — and which are merely convenient?
- ? What is lost in the measurement?

Designing the System

MODEL AND OBJECTIVE

- ? What does the model family assume, and how flexible is it?
- ? Does minimising this loss correspond to what we actually want?

LEARNING

- ? Which procedure searches the parameters?
- ? How sensitive is the result to initialisation and randomness?

Designing the System — Evaluation

EVALUATION AND GENERALISATION

- ? Which metric matters, and which data is held out?
- ? How uncertain is the estimate of performance?
- ? What happens when the environment changes?

UNCERTAINTY AND USE

- ? Which uncertainty must be represented explicitly?
- ? Is the prediction informing a decision — and what do the errors cost?

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These are the questions the rest of the course answers. Each lecture takes one or two of them and makes them precise.

Takeaway

> Before any model is fitted, the design questions are answered — what one prediction is, which data stands for the world, what counts as success. Each answer is an **assumption**; together they decide whether any model can succeed.

Module 2

How Learning Is Possible

> ***Guiding Question:*** *Infinitely many rules fit the data we have seen. What licenses a machine to prefer one of them?*

Learning as Search

The **hypothesis space** $\mathcal{H}$ is the set of all functions the model family can express — every possible value of the parameters Θ. Learning is the search for the best $h_\Theta \in \mathcal{H}$.

Exhaustive search

Examine all of $\mathcal{H}$. Feasible only when the space is small (with a few special exceptions).

Heuristic search

Use assumptions to explore only part of the space. Efficient, but the optimum may be missed.

Except for very simple problems you cannot have both: searching **all** hypotheses **efficiently and robustly**.

Inductive Reasoning and Inductive Bias

Inductive reasoning: generalising from seen examples to unseen ones — from observations to a rule that predicts future observations.

Inductive bias: the assumptions by which an algorithm generalises. The choice of hypothesis space is already one; so are assumptions about noise, outliers, the data distribution and the shape of errors.

> Without an inductive bias, a system **cannot generalise** at all.

The Bias–Variance Trade-Off

Learning trades **fitting the observed data** against **holding to prior assumptions about unseen data**.

Bias

Systematic error: averaged over many training sets, the model still misses the truth, because its assumptions rule it out — the *stubborn learner*.
High bias → robust to noise, but misses real structure → **underfitting**.

Variance

How much the learned model changes from one training set to the next — it follows irregularities of the particular sample.
High variance → captures patterns, but also noise → **overfitting**.

> More parameters generally means more variance. The trade-off is not a defect of particular algorithms — it is a property of inductive inference.

Takeaway

> Learning is a **search** through a hypothesis space, steered by an **inductive bias** you chose — knowingly or not. Every model sits somewhere on the bias–variance axis, and no model escapes it.

Module 3

Linear Regression

> ***Guiding Question:*** *Why squared error — and what exactly does "linear" forbid?*

Classification versus Regression

Classification

Y is a finite set of labels.

Will this customer default — yes or no?

Regression

$Y = \mathbb{R}$, the target is a quantity.

What credit line should this customer receive?

The learning machinery is the same; only the shape of Y — and therefore the loss — changes.

The Model

Linear regression predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + \ldots + w_p x_p = \mathbf{w}^\top \mathbf{x}$$

with $x_0 = 1$ absorbing the intercept.

The hypothesis space is $\mathbb{R}^{p+1}$: one hypothesis per weight vector.

The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w} \in \mathbb{R}^{p+1}} \mathrm{RSS}(\mathbf{w}), \qquad \mathrm{RSS}(\mathbf{w}) = \sum_{i=1}^{n} \left(y_i - \mathbf{w}^\top \mathbf{x}_i\right)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sum_i (x_i - \overline{x})^2}$$

The covariance of x and y over the variance of x.

Intercept

$$\hat{w}_0 = \overline{y} - \hat{w}_1 \overline{x}$$

The fitted line passes through the point of means.

How Much Variance Is Explained?

$$\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}, \qquad R^2 = \frac{\mathrm{ESS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares (the split holds for a least-squares fit with an intercept).

R^2 is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

Training R^2 never decreases when a feature is added, however useless the feature. A high training R^2 is not evidence that a model generalises — test R^2 and adjusted R^2 can and do fall.

Higher Dimensions

With p features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever $\mathbf{X}^\top \mathbf{X}$ is invertible.

> Invertibility fails when features are linearly dependent or when $p > n$ — more unknowns than equations. Both are common in practice; regularisation is the standard remedy.

Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and R^2 reports how much variation it explains — not whether it will generalise.

Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights** $\mathbf{w}$. The features may be arbitrary transformations of the input.

Plain

$$h(\mathbf{x}) = w_0 + w_1 x$$

a straight line

Polynomial

$$h(\mathbf{x}) = w_0 + w_1 x + w_2 x^2$$

a curve — still a *linear model*

Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map ϕ

What *linear* forbids: a transformation that has to be learned from the data. The feature map is fixed before fitting; only the weights are learned.

Consequences

The good

The closed-form solution and all statistical guarantees carry over unchanged.

The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits; too poor and it underfits.

> Increasing the polynomial degree turns a dial on the bias–variance axis.

The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

Geometric View

The predictions $\mathbf{Xw}$ live in the **column space** of $\mathbf{X}$ — everything the features can express. The observed $\mathbf{y}$ generally does not.

Least squares picks the point of that space **closest to $\mathbf{y}$ in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

Consequence

The residual carries no remaining linear information about the features.

The Gauss–Markov Theorem

Gauss–Markov (informal). Among all estimators that are

- **linear** in the observed outputs $(y_1, \ldots, y_n)$, and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

Best

lowest variance among the candidates

Linear

a linear function of the observations

Unbiased

right on average over repeated samples

What This Buys, and What It Costs

The payoff

Minimising RSS yields optimal statistical properties under **mild assumptions** — no normality needed, only a correctly specified linear model, linearly independent features, and zero-mean, uncorrelated errors of constant variance.

The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

Fitting h by Repeated Correction

LMS (least mean squares)

1. Initialise the weights w.
2. Pick a training example $(\mathbf{x}, y)$ from the training set D.
3. Compute the prediction $h_w(\mathbf{x})$ and the error $e = y - h_w(\mathbf{x})$.
4. Update every weight: $w_i \leftarrow w_i + \eta e x_i$.
5. Repeat 2–4 until the error on D stops decreasing.

What the Update Rule Encodes

Direction

The sign of e decides whether the prediction goes up or down.

Attribution

The factor x_i gives each feature credit proportional to its value.

Step size

The learning rate η trades speed against stability.

Too large an η oscillates or diverges; too small an η converges slowly. It is a hyperparameter — chosen, not learned.

Two Ways to the Same Minimum

Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a $(p+1) \times (p+1)$ matrix — costly for large p, impossible when it is singular.

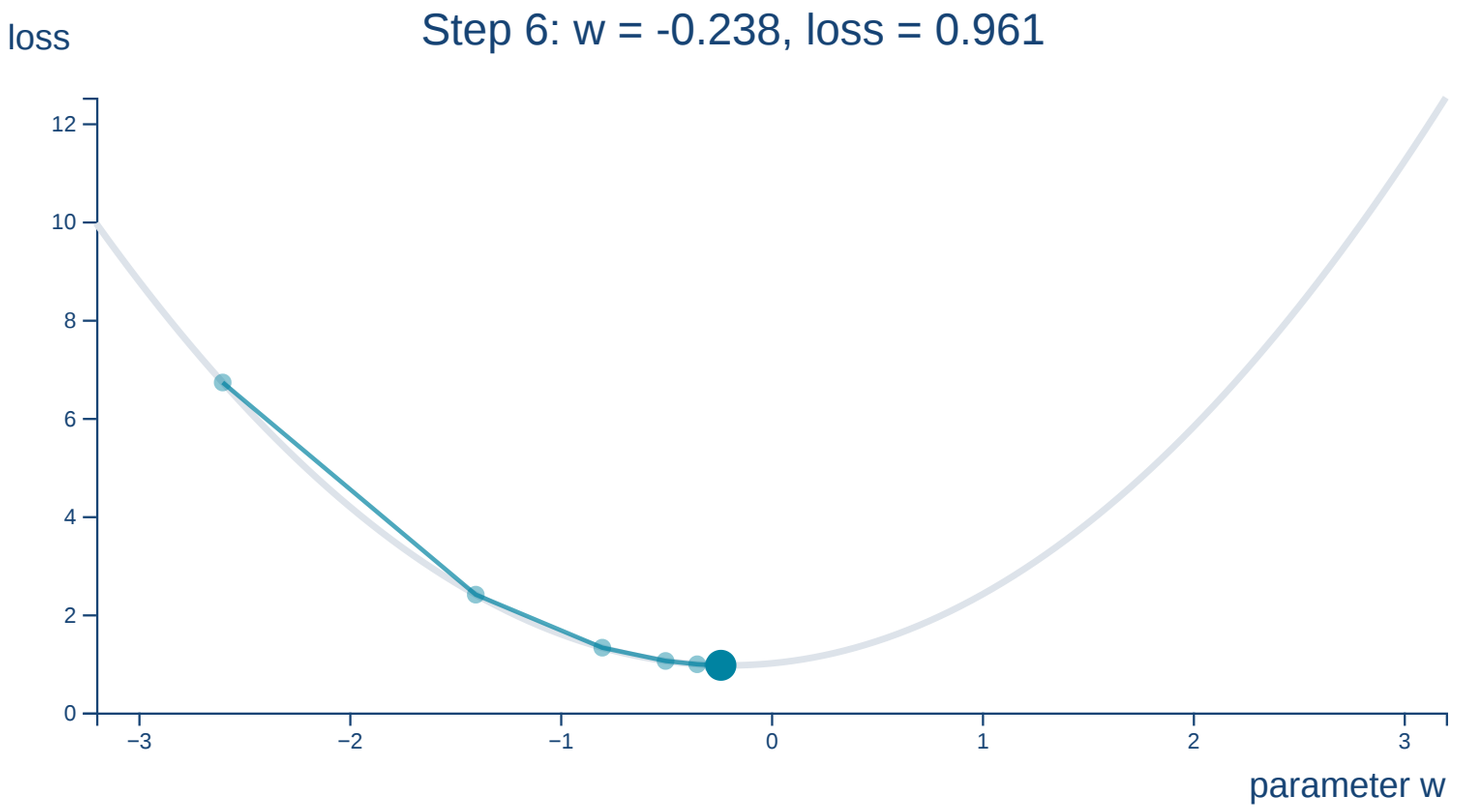
Iterative (gradient descent)

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \mathrm{RSS}(\mathbf{w})$$

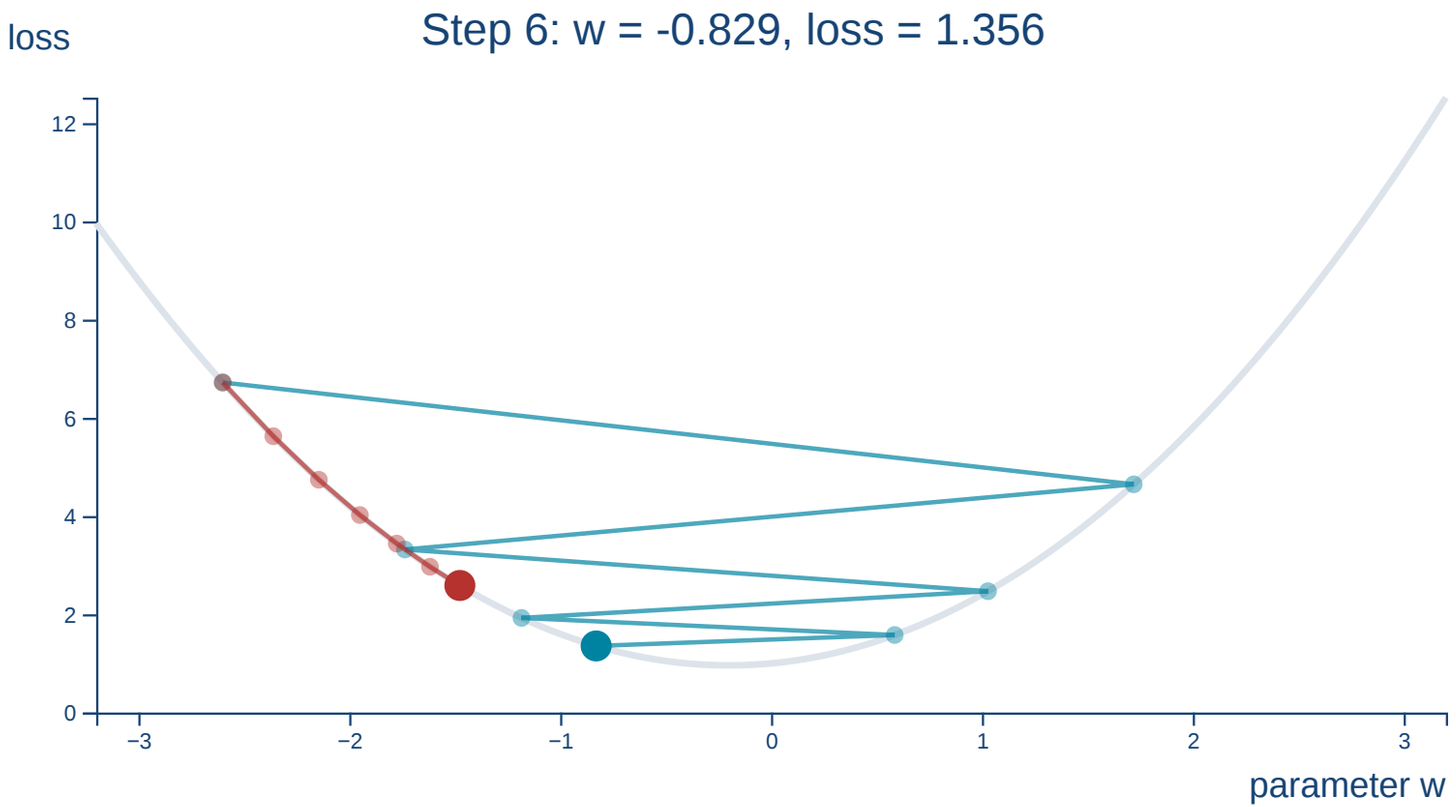
Approximate, many steps. Scales to large p and n, works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm**: its update $w_i \leftarrow w_i + \eta e x_i$ *is* a gradient step on RSS, taken per example instead of per dataset (the factor 2 is absorbed into η).

Watching Gradient Descent Work



The Learning Rate Decides Everything



Teal: $\eta = 0.9$ — overshoots and oscillates across the minimum. Red: $\eta = 0.05$ — stable, but slow: after six steps it has covered about half the distance.

The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

Takeaway

> Use the **closed form** when the matrix is small and well-conditioned; use **gradient descent** when it is not — and for most models beyond linear regression, no closed form exists at all.

Module 4

From Regression to Classification

> ***Guiding Question:*** *A decision is a yes or a no. What goes wrong when we fit a straight line to one?*

Fitting a Line to Labels

Encode the two classes as $y \in \{-1, +1\}$, fit a linear model, and classify by the sign of the prediction:

$$h(\mathbf{x}) = \operatorname{sign}\left(\mathbf{w}^\top \mathbf{x}\right)$$

One feature

The **discrimination point** x' satisfies

$$w_0 + w_1 x' = 0$$

Everything to one side is one class.

Two features

The **discrimination line** satisfies

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

In p dimensions it becomes a hyperplane.

Where It Breaks

Squared loss punishes correctness

A point far on the *correct* side has a large residual, so least squares drags the boundary toward it.

The boundary shifts

Adding well-classified points — or an outlier — moves the discrimination point, even when nothing was misclassified.

The output is also unbounded: $\mathbf{w}^\top \mathbf{x}$ can be 17 or −40, and neither is interpretable as a degree of belief.

Two Problems, One Fix

$\mathrm{sign}(\mathbf{w}^\top \mathbf{x})$ is not differentiable

No gradient means no gradient descent — the optimisation machinery does not apply.

The boundary is unstable

Squared loss on ± 1 labels lets well-classified points shift the discrimination point.

Replace the hard threshold with the **logistic sigmoid**:

$$h_\Theta(\mathbf{x}) = \mathrm{expit}\left(\mathbf{w}^\top \mathbf{x}\right) \approx P(y = 1 \mid \mathbf{x})$$

The output becomes a smooth, bounded quantity that can be read as a probability — and can be differentiated.

Turning a Model Into an Estimator

We have a model with parameters θ and data D. The model assigns each possible dataset a probability.

The **likelihood** is that probability, read as a function of the **parameters** with the data held fixed:

$$L(\theta) = p(D \mid \theta)$$

The **maximum likelihood estimate** is $\hat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(D \mid \theta)$.

$p(D \mid \theta)$ is a probability over **data**, not over parameters. The likelihood is not a distribution over θ and does not integrate to 1.

Why the Logarithm

Under the **i.i.d. assumption** — observations independent and identically distributed — the likelihood factorises:

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta) \qquad \Longrightarrow \qquad \ell(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

Same maximum

$\log$ is monotone, so it does not move the arg max.

Numerically safe

A product of thousands of small probabilities underflows; a sum of logs does not.

Differentiable sums

Sums differentiate term by term — gradient methods apply directly.

> The i.i.d. assumption is doing real work here. Time series, spatial data and grouped data violate it, and the factorisation is then simply wrong.

Example: A Biased Coin

n flips, h heads, parameter $\theta = P(\text{heads})$:

$$\ell(\theta) = h \log \theta + (n - h) \log(1 - \theta) \qquad \frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \qquad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

The estimator is the observed frequency — the answer intuition would have given. MLE **derives** it rather than assuming it, and the same derivation works where intuition has nothing to offer.

Mini-Exercise

You flip a coin **3 times** and see 3 heads. What is $\hat{\theta}_{\mathrm{ML}}$ — and would you bet that this coin never shows tails?

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You flip a coin **3 times** and see 3 heads. What is $\hat{\theta}_{\mathrm{ML}}$ — and would you bet that this coin never shows tails?

$\hat{\theta}_{\mathrm{ML}} = 3/3 = 1$: the estimate rules tails out. Three flips are no basis for that. The estimator uses no **prior** knowledge (coins are usually near-fair) and reports no **uncertainty** — a single point, no error bar. A prior and a posterior close both gaps.

Takeaway

Maximum likelihood picks the parameters under which the observed data is most probable. It is the bridge from a **model of the world** to a **learning algorithm** — and it is blind to prior knowledge and to its own uncertainty.

Odds and Log-Odds

Odds

The ratio of favourable to remaining cases.

Rolling a six: odds 1 : 5.

Logit

The **log-odds** of a probability p:

$$\operatorname{logit}(p) = \ln \frac{p}{1-p}$$

Range

$p \in (0, 1)$ maps to $(-\infty, +\infty)$ — exactly the range a linear model produces.

The modelling assumption: the logit of the target is a linear combination of the features.

$$\ln \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}$$

The Model

Inverting the logit gives the **logistic sigmoid**:

$$h_\Theta(\mathbf{x}) = \operatorname{expit}(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

Shape

An S-curve: near-linear around 0, saturating towards 0 and 1 at the extremes.

Reading

$h_\Theta(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$, and the decision boundary is still $\mathbf{w}^\top \mathbf{x} = 0$ — where the sigmoid passes 0.5.

The Loss: Cross-Entropy

Binary cross-entropy (log loss)

$$L(\Theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \ln h_\Theta(\mathbf{x}_i) + (1-y_i)\ln\left(1-h_\Theta(\mathbf{x}_i)\right)\right]$$

For $|Y| > 2$ classes, the multiclass form sums over the classes with a softmax output.

Where does it come from? From the **likelihood**, with the labels now coded as $y_i \in \{0, 1\}$: $P(y_i = 1 \mid \mathbf{x}_i; \Theta) = h_\Theta(\mathbf{x}_i)$ and $P(y_i = 0 \mid \mathbf{x}_i; \Theta) = 1 - h_\Theta(\mathbf{x}_i)$. Maximising the likelihood of the data is minimising this loss.

Why It Fixes the Problem

Confident and correct

Loss goes to zero — no penalty for being emphatically right.

Confident and wrong

Loss goes to infinity — exactly the errors we want punished.

There is no closed-form solution; the parameters are found by gradient descent on this loss.

Takeaway

Logistic regression is a **linear model of the log-odds**, trained by maximising likelihood. It keeps the linear decision boundary but produces a probability — one whose calibration can be checked — and a loss that punishes the right kind of error.

Module 5

Probabilistic Classification

> ***Guiding Question:*** *The data tells us how attributes behave given the class. We need the reverse. What does the conversion cost?*

Two Directions of the Same Connection

Observed connection

$$A_1, \ldots, A_p \mid B$$

Given the class, how do the attributes behave? — the **generative** direction: modelled class by class.

Reversed connection

$$B \mid A_1, \ldots, A_p$$

Given the attributes, which class? — the diagnosis situation, and the one we actually need.

$$P(B \mid A_1, \ldots, A_p) = \frac{P(B) \cdot P(A_1, \ldots, A_p \mid B)}{P(A_1, \ldots, A_p)}$$

What the Classifier Needs

Class priors

$$P(\text{class} = y)$$

How common is each class, before seeing any attribute?

Attribute likelihoods

$$P(\text{attribute} = v \mid \text{class} = y)$$

How does each class distribute its attribute values?

Given both, the posterior $P(\text{class} \mid \text{attributes})$ follows from Bayes — and the prediction is the class with the highest posterior.

> Estimating $P(A_1, \ldots, A_p \mid B)$ requires the **joint** distribution of all attributes within each class: exponentially many parameters, and never enough data.

Generative and Discriminative Classifiers

Generative

Model the **joint** distribution $P(A, B)$ — how the data was produced — then derive $P(B \mid A)$.

Naive Bayes.

Discriminative

Model $P(B \mid A)$ **directly** — only the boundary between classes.

Logistic regression.

The Naive Assumption

Assume the attributes are **conditionally independent given the class**:

$$P(A_1, \ldots, A_p \mid B) = \prod_{j=1}^{p} P(A_j \mid B)$$

The full joint needs $2^p - 1$ parameters per class for p binary attributes; under the assumption it needs one distribution per attribute — p numbers per class.

The assumption is almost always false. Naive Bayes nonetheless classifies well, because ranking the classes correctly is easier than estimating their probabilities correctly.

The Classifier

$$P(B_i \mid A_1, \ldots, A_p) = \frac{P(B_i) \cdot \prod_{j=1}^{p} P(A_j \mid B_i)}{\sum_{l=1}^{k} P(B_l) \cdot \prod_{j=1}^{p} P(A_j \mid B_l)}$$

Score each class

prior × product of attribute likelihoods

Normalise

divide by the total over all classes — the theorem of total probability

Predict

the class with the largest posterior

For prediction alone the denominator can be skipped — it is the same for every class. It is needed only when the probability itself is reported.

Example: Enjoy Sport?

Outlook	Temperature	Humidity	Wind	EnjoySport
sunny	hot	high	weak	no
sunny	hot	high	strong	no
overcast	hot	high	weak	yes
rain	mild	high	weak	yes
rain	cool	normal	weak	yes
rain	cool	normal	strong	no

Estimate the priors and the per-attribute likelihoods by **counting**, then score a new day: rain, hot, high, weak.



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Estimate the priors and the per-attribute likelihoods by **counting**, then score a new day: rain, hot, high, weak.

$$\text{yes: } \tfrac{1}{2} \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} \cdot \tfrac{2}{3} \cdot 1 \approx 0.074 \qquad \text{no: } \tfrac{1}{2} \cdot \tfrac{1}{3} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} \approx 0.025$$

$P(\text{yes} \mid \text{day}) = 0.074/(0.074 + 0.025) = 0.75$ — predict **yes**.

The Zero-Frequency Problem

If an attribute value never occurs with a class in the training data, its estimated likelihood is **zero** — and one zero factor drives the whole product to zero, whatever the other attributes say.

Laplace smoothing adds a pseudo-count to every combination, so that unseen does not mean impossible:

$$\hat{P}(A_j = v \mid B) = \frac{n(v, B) + 1}{n(B) + k_j}$$

with k_j the number of values of attribute A_j.

One missing count

vetoes every other piece of evidence

Naive Bayes buys tractability with a **conditional independence assumption it knows to be false** — and works anyway, because classification needs the right argmax, not the right probability.

Module 6

Measuring Performance

> ***Guiding Question:*** *A model reports 99.9% accuracy and has never detected a single case. What should we have measured instead?*

What We Want to Know

> The **true misclassification rate** of a hypothesis h is the probability that it errs on a randomly drawn example $(X, Y) \sim p$:
>
> $$\mathrm{err}^*(h) = P\big(h(X) \neq Y\big)$$

It is defined over the **whole distribution of examples** — which we cannot observe. Since the same $\mathbf{x}$ can come with different labels, even the best possible h errs sometimes: that floor is the **Bayes error**.

What We Can Measure

We estimate $\mathrm{err}^*(h)$ by the error on a set of examples $D_s \subseteq D$ whose true labels are known:

$$\mathrm{err}(h, D_s) = \frac{1}{|D_s|} \sum_{(\mathbf{x},y)\in D_s} 1\big[h(\mathbf{x}) \neq y\big]$$

> The estimate is only as good as the sample. Which examples land in D_s — and whether h has seen them before — decides whether the number means anything.

Training Error Is Not an Honest Estimate

When the evaluation set equals the training set, $D_{tr} = D$, the resulting number measures how well the model **memorised**, not how well it will generalise.

> Training error is optimistically biased — typically below the true error, often far below. A model with enough capacity can drive it to zero while learning nothing.

Four Estimation Schemes

Scheme	How	Trade-off
Hold-out	split D once into a training and a test part; fit on one, evaluate on the other	simple, but wastes data and depends on the split
k-fold cross-validation	partition D into k folds; each fold serves once as the test set; average the k errors	uses all data for both roles; k = 5 or 10 is customary
Leave-one-out	$k = n$: every example is a test set of size one	nearly unbiased, expensive, high variance
Bootstrapping	resample D with replacement to build training sets; evaluate on the examples left out	useful when D is small

Misclassification Costs

Not every error costs the same. A missed diagnosis and a false alarm have different consequences, so counting errors equally is itself a modelling assumption.

> A **cost matrix** assigns a cost to each (true class, predicted class) pair; the objective becomes expected cost rather than error rate.

Takeaway

> An error number is meaningless without the sampling scheme that produced it. **Never evaluate on data the model was fitted on**, and state the cost model when errors are not interchangeable.

A Number Alone Means Nothing

"95% accuracy" is not a result until you know what a trivial method achieves.

Trivial classification

Always predict the **majority class**.

Trivial regression

Always predict a **central tendency** — the mean or median of y.

Random guessing

Draw a label from the marginal distribution $p(y)$.

Example: Why Accuracy Deceives

A screening test for a disease affecting **1 in 1,000** people.

The "model"

Always predict *healthy*.

Its accuracy

99.9% — and it has never detected a single case.

> On imbalanced data the majority baseline is nearly unbeatable by accuracy, and completely useless. The metric hides that the model is useless.

Why the Error Rate Is Not Enough

For multiclass classification $h_\Theta : X \to Y$ with $Y \subseteq \{1, \ldots, N\}$, the plain error rate:

Fails on imbalance

The majority baseline scores high while detecting nothing.

Hides the distribution

Which classes the errors fall on stays invisible.

Hides the kind of error

Missing assignments and imprecise assignments count the same.

The Confusion Matrix

Rows are true classes, columns are predictions; cell (i, j) counts the examples of class i that h assigned to j. The diagonal is correct, and every off-diagonal cell names a specific confusion.

> The matrix is the raw material: every measure below is a ratio of its cells. Read it before reading any summary number.

Four Counts, Per Class

For a class j, every prediction falls into one of four cells:

	truly in j	**not in j**
assigned to j	TP_j — true positives	FP_j — false positives
not assigned to j	FN_j — false negatives	TN_j — true negatives

> Read the letters as **decision then correctness**: P/N is what h decided, T/F is whether that decision was right.

Precision and Recall

Recall — completeness

$$\mathrm{rec}_j = \frac{TP_j}{TP_j + FN_j}$$

Of everything that truly belongs to j, what fraction did h find?

Precision — exactness

$$\mathrm{prec}_j = \frac{TP_j}{TP_j + FP_j}$$

When h says j, how often is it right?

Both are **conditional probabilities with different conditions** — recall conditions on the truth, precision on the prediction. Swapping them is the same error as swapping the conditionals in Bayes' theorem.

One number that is high only if both are: their harmonic mean, $F_1 = \frac{2 \cdot \mathrm{prec}_j \cdot \mathrm{rec}_j}{\mathrm{prec}_j + \mathrm{rec}_j}$.

Mini-Exercise

A spam filter flags 80 of 1,000 emails; 60 of them are spam, out of 100 spam emails in total. What are precision and recall — and which would you rather improve?



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A spam filter flags 80 of 1,000 emails; 60 of them are spam, out of 100 spam emails in total. What are precision and recall — and which would you rather improve?

- **Precision** $= 60/80 = 0.75$: a quarter of flagged mail is legitimate.
- **Recall** $= 60/100 = 0.60$: forty spam messages still arrive.

For a spam filter, **precision** usually matters more: a lost legitimate email costs far more than a spam message that slips through. The cost matrix decides, not the metric.

Takeaway

> The confusion matrix shows **which** errors occur; precision and recall split "being wrong" into **being imprecise** and **being incomplete**. Which one you optimise is a decision about costs, not about mathematics.

The Churn Shop: Accuracy or Recall?

Of **1,000** customers, **30** will churn in the next quarter (3%). Two models:

	flags	churners caught	accuracy	recall	precision
"nobody churns"	0	0 of 30	97.0%	0%	—
churn alert	60	24 of 30	95.8%	80%	40%

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Which model is better depends on the **costs**: a retention call for 60 customers against 24 kept.

Squared Error

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(h(\mathbf{x}_i) - y_i\right)^2 \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}$$

MSE

Average quadratic loss; always non-negative, lower is better.
Unit is the **square** of the target's unit — hard to interpret directly.
Large errors are penalised disproportionately.

RMSE

Same ordering as MSE, but back in the **target's own unit** — which is why it is the one usually reported.

Absolute and Relative Error

MAE

$$\frac{1}{n}\sum_i \left|h(\mathbf{x}_i) - y_i\right|$$

Easier to interpret than RMSE; treats all errors proportionally.

Mean error

Drop the absolute value to keep the **direction** — reveals systematic over- or under-prediction; errors of opposite sign cancel.

Mean relative error

$$\frac{1}{n}\sum_i \frac{\left|h(\mathbf{x}_i) - y_i\right|}{\left|y_i\right|}$$

The average error *per unit of target* — comparable across scales, undefined where $y_i = 0$.

Normalised RMSE variants aim to make errors comparable between datasets, but are hard to interpret. Prefer a measure whose unit you can name.

Takeaway

Squared measures punish large errors and lose the unit; absolute measures stay interpretable; signed measures expose bias. Report the one whose **unit and failure mode** match the decision the prediction feeds.